

Darren Trieu

darrentrieu@gmail.com - (858) 429-8209

Education

University of California, San Diego <i>M.S Computer Science</i> GPA 3.79	Class of 2024
Coursework: Probabilistic Reasoning & Learning, Databases, Programming Languages, Algorithms, Networking and Distributed Programming, Embedded Systems, Data Mining, Recommender Systems	
University of California, San Diego <i>B.S Pharmacological Chemistry</i> GPA 3.77	Class of 2022

Experience

Software Engineer - General Atomics	Apr. 2025 - Present
--	---------------------

- Own a high-throughput distributed logging service processing **1000+ Hz** message traffic with **zero data loss** and minimal system overhead.
- Improve memory usage and logging throughput by **500%+** by redesigning storage from per-message disk writes to asynchronous in-memory buffering.
- Lead development of a **mission-critical operational control platform** with a React frontend and **C++ backend** supporting real-time distributed communication during live flights.
- Drive **cross-team integration** and feature development for the platform while mentoring **2 junior engineers** and coordinating requirements across multiple engineering teams.
- Lead flight integration efforts as the **primary point of contact** for the Mission Systems software kit, coordinating networking and **real-time data exchange** across distributed systems to ensure flight success.
- Contribute to **containerized CI/CD workflows** using Podman and Kubernetes, ensuring code reliability through static analysis, automated testing, build validation, and cluster health monitoring.
- Integrate **AI-assisted coding agents** using Claude Code with internal API and documentation context to accelerate feature development and debugging

Software Engineering Intern - UCSD Weibel Lab	Jun. 2024 - Apr. 2025
--	-----------------------

- Designed MongoDB schemas for vector embeddings and integrated **MongoDB Atlas Vector Search**, utilizing **Ollama-hosted local models** for embedding generation and experimentation workflows.
- Developed backend services using **Express** and **Node.js** to process medical device data and support real-time client-server communication.
- Built RESTful APIs for audio processing and AI interaction using Deepgram and OpenAI integrations, implementing a **state-machine workflow** to optimize token usage through conversation summarization.
- Developed a **clinical decision-support algorithm** recommending medical interventions and care plans using real-time patient interaction data.

Projects

SurfStore Cloud Storage Project - Go, gRPC, Protocol Buffers	Mar. 2024 - Jun. 2024
---	-----------------------

- Designed and implemented a **cloud file storage service** that supported file synchronization and retrieval across distributed systems.
- Engineered SurfStore's **distributed storage layer**, utilizing a consistent hashing-based BlockStore for efficient data placement/retrieval and a RAFT-replicated MetaStore ensuring consistency and fault tolerance.

Skills

Languages: Python, Java, C++, JavaScript, HTML/CSS, SQL

Technologies: React, Node.js, Express, MongoDB, OpenAI API, Deepgram, Vector Search, Kubernetes, Podman

Workflow/Tools: Git, GitHub, CI/CD, Agile, Jira, Jama